

Making sense of time-travel: Dissociation between co-speech gesture and temporal reasoning

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Introduction

- People **produce gestures spontaneously** while talking about temporal relations (Cooperrider & Núñez 2009; Casasanto & Jasmin 2012; Walker & Cooperrider 2016)
- Time travel is an incredible case of conceptual complexity, and a model system of abstract temporal reasoning
- On some accounts, gesture makes a **causal contribution** to abstract mathematical reasoning (e.g., Cook et al. 2008); however, it remains unclear whether **temporal gestures** play a facilitative role in temporal reasoning
- Alternatively, gestures may reflect internal cognitive processes, but the gestures **may not contribute causally** to abstract reasoning (Walkington et al., 2019)

Research Question

Is temporal gesture **necessary** in temporal reasoning?

Methods

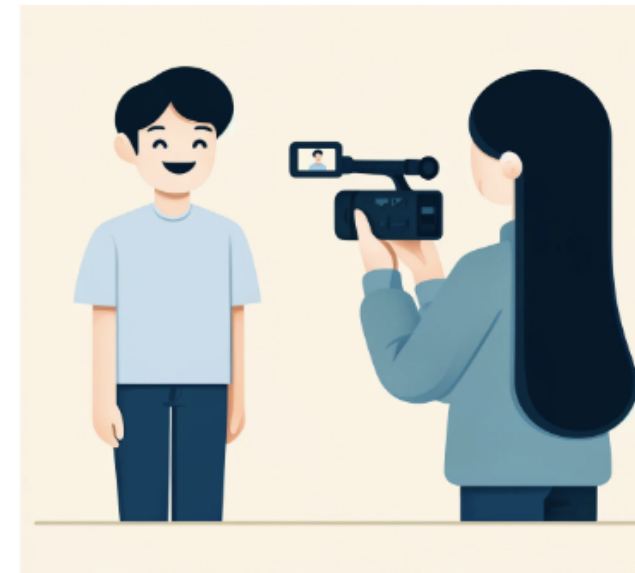
Participants:

- $N = 44$ college students at an American university
- The study was conducted in English
- Participants' native language and country of origin varied

Design:



Participants watched short videos



Video-taped narrative retelling



Reasoning questions on computer



time-travel narratives:
The Simpsons - Treehouse of Horror Tethers (2021)
non time-travel narrative:
Seinfeld

No explicit gesture elicitation

Asked targeted temporal reasoning questions

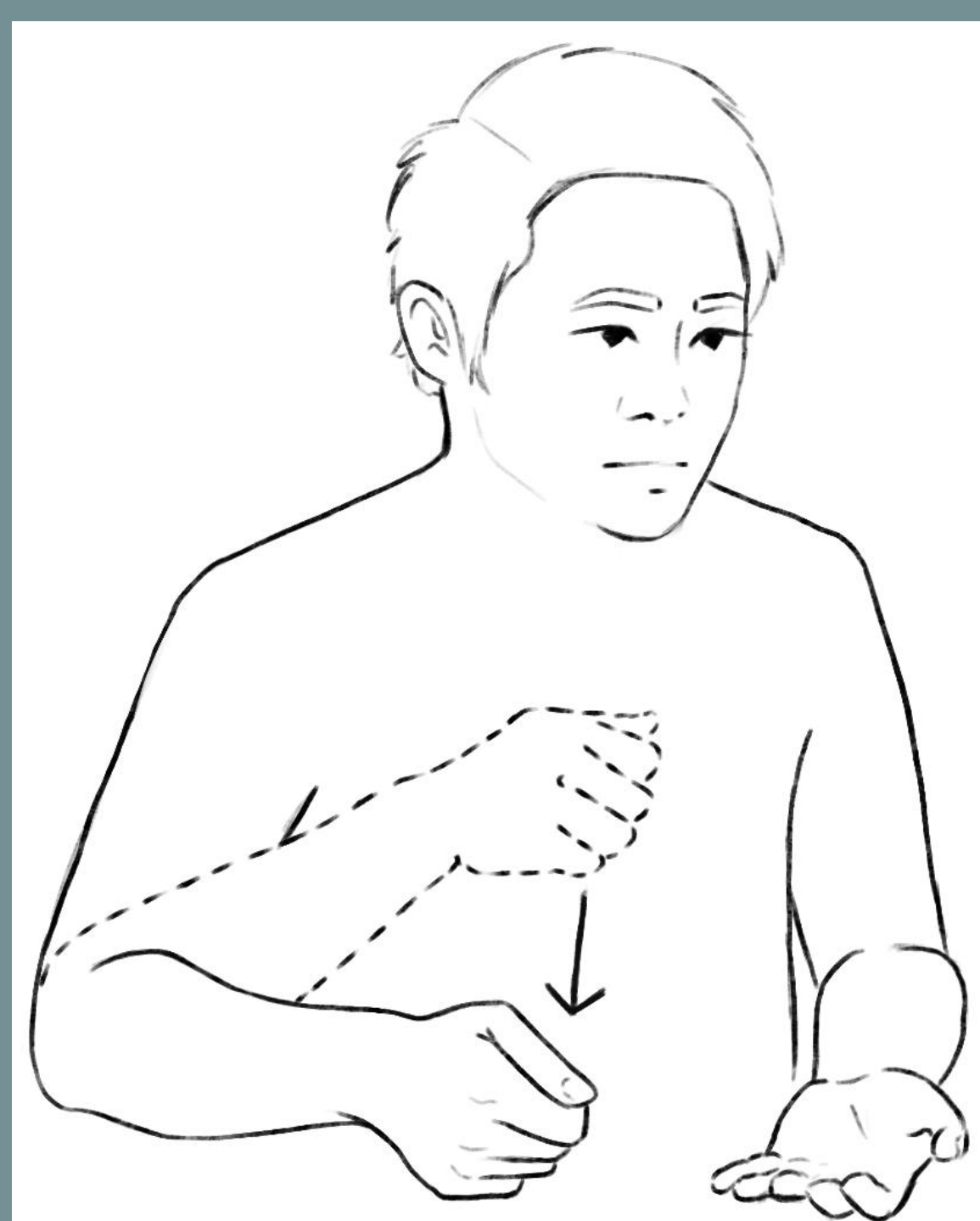
Sequence Simpsons story from 2 different perspectives

8X Recall questions

12X **Temporal sequencing** questions

Analysis:

- Temporal sequencing question responses as seen from two different perspectives were analyzed
- For temporal gestures, manual gestures were coded that depicted the temporal content of the language in speech
- Representational gestures (iconic and non-temporal metaphoric)



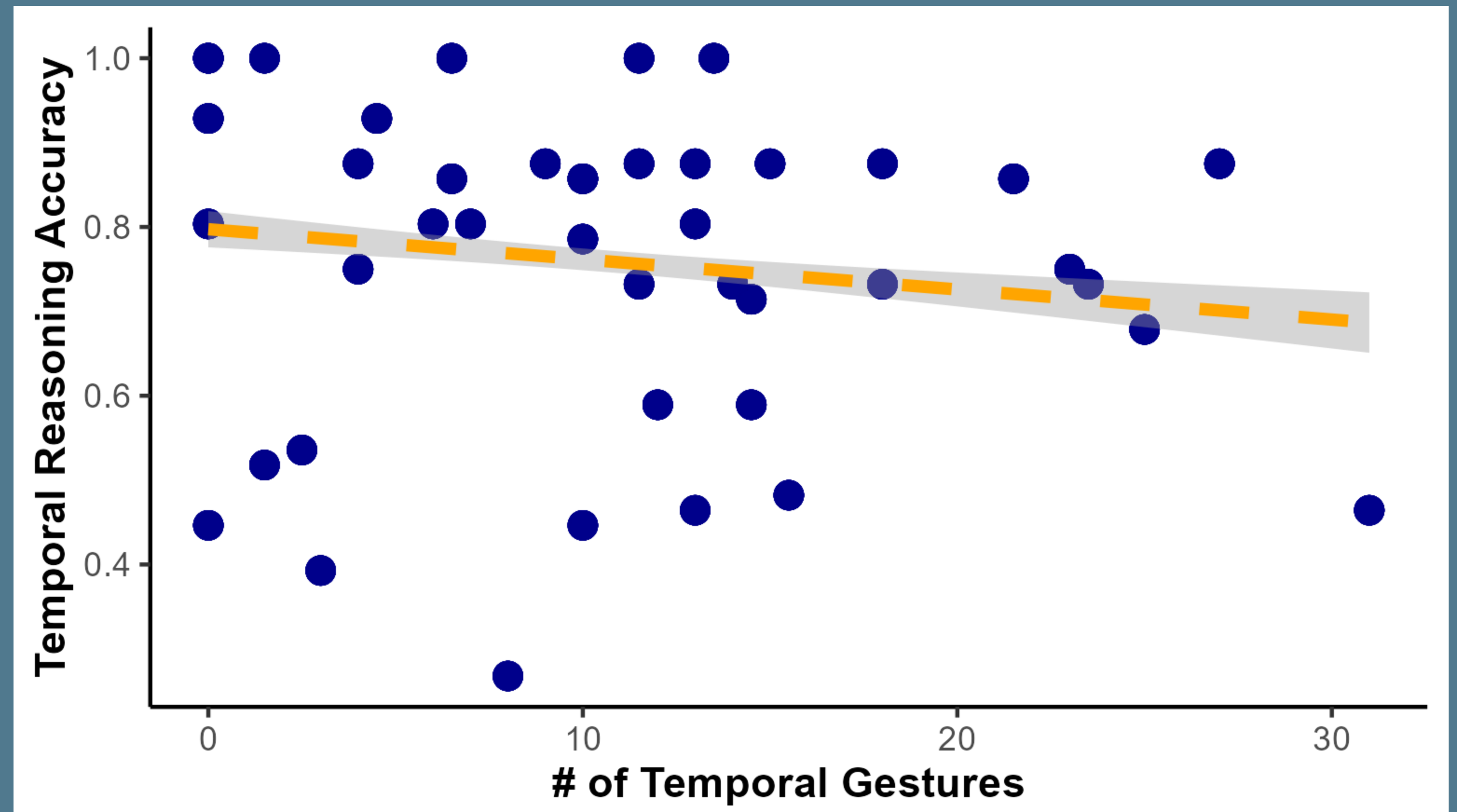
Representational



Temporal

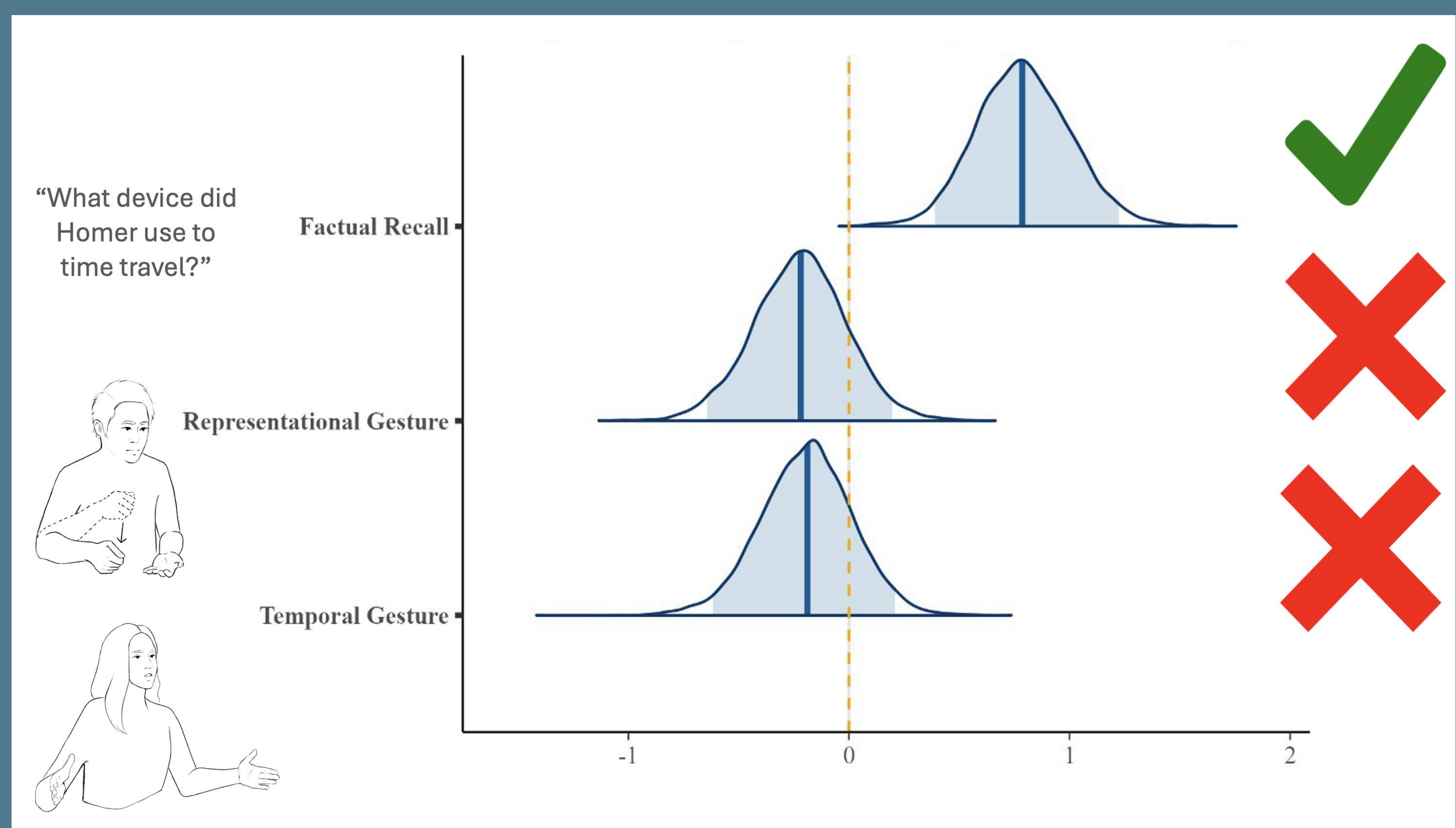
Results

Association between temporal gesture and temporal reasoning accuracy



Negative correlation between the number of temporal gestures produced and temporal reasoning accuracy
Pearson's $r = -0.147$ $p < .001$

Predicting temporal reasoning accuracy



Posterior distributions from multilevel Bayesian model (z-scored predictors)
Factual recall predicts temporal reasoning, but representational and crucially, temporal gesture, **do not**

Discussion

Radical accounts of embodied cognition:

- The absence of an association between gesture and abstract reasoning could indicate a **lack of necessary causal function**

Facilitative role of gesture in abstract reasoning:

- Individuals without stable spatiotemporal schemas may need to gesture more
- Bodily affordances are **redundant and unneeded** for those with already stable spatiotemporal schemas

Differences between gesture and abstract reasoning about Math vs Time:

- If gesturing supports learning of abstract concepts in mathematics, then why does it not support **temporal** reasoning?
 - Spontaneous vs. experimentally manipulated gesturing
 - These time travel narratives may lack sufficient complexity

Conclusion

- Temporal gesture frequency **does not predict** temporal reasoning ability
- Temporal gesture is **not necessary** in temporal reasoning

References

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- Cook, Susan Wagner, Zachary Mitchell & Susan Goldin-Meadow. 2008. Gesturing makes learning last. *Cognition* 106(2). 1047–1058. doi:10.1016/j.cognition.2007.04.010.
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